TR1 Worked Examples: Degrees, Radians and Standard Position

Recording companion for the TR1 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

How much of a half turn is this angle? A half turn is 180 degrees and also π radians, so 180° = π. To go from degrees to radians, multiply by π/180; from radians to degrees, multiply by 180/π. To draw an angle in standard position, put its vertex at the origin, start on the positive x-axis, and turn counterclockwise for a positive angle.

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Example 1: 150 Degrees

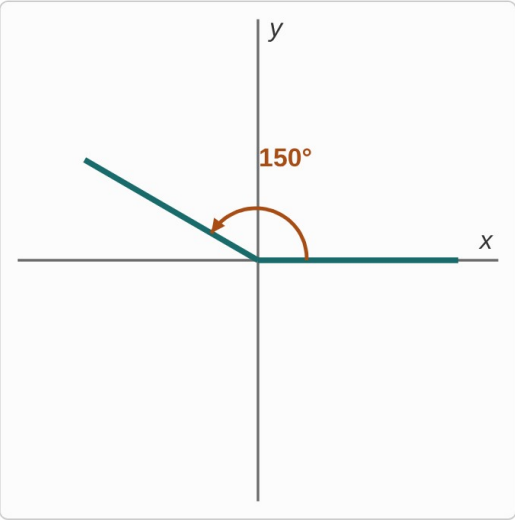


Figure: An angle in standard position: the initial side lies on the positive x-axis, and an arc labeled 150° turns counterclockwise to the terminal side, which points up and to the left, 30° short of the negative x-axis.

$$150°$$

Question

What is this angle in radians?

- **A.** $\pi/6$
- **B.** $5\pi/6$
- **C.** $7\pi/6$
- **D.** $5\pi/3$

Answer and reasoning

$150° \cdot \pi/180 = 150\pi/180 = 5\pi/6$, dividing top and bottom by 30. A check: 150° is a little less than 180°, and $(5/6)\pi$ is a little less than π. $\pi/6$ is 30°, the reference angle, not the angle itself; $7\pi/6$ is 210°, the mirror image of 150° across the x-axis; and $5\pi/3$ divides by 90 instead of 180, as if π were a quarter turn. ***5π/6*.**

Example 2: 225 Degrees

$$225°$$

Question

What is this angle in radians?

A. $5\pi/4$
B. $\pi/4$
C. $3\pi/4$
D. $7\pi/4$

Answer and reasoning

$225 \cdot \pi/180 = 225\pi/180 = 5\pi/4$, dividing by 45. 225° is 45° past 180°, and $5\pi/4$ is $\pi/4$ past π. $\pi/4$ is 45°, the reference angle; $3\pi/4$ is 135°, the mirror image across the x-axis, in Quadrant II; and $7\pi/4$ is 315°, the mirror image across the y-axis, in Quadrant IV. ***5π/4*.**

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Example 3: 45 Degrees

$$45°$$

Question

What is this angle in radians?

A. $\pi/2$
B. $3\pi/4$
C. $7\pi/4$
D. $\pi/4$

Answer and reasoning

$45 \cdot \pi/180 = 45\pi/180 = \pi/4$, dividing by 45: an eighth of a full turn. $\pi/2$ divides by 90 instead of 180, as if π were a quarter turn; $3\pi/4$ is $180° - 45°$, the mirror image in Quadrant II; and $7\pi/4$ is $360° - 45°$, which is coterminal with $-45°$, not with $45°$. ***π/4*.**

Example 4: 330 Degrees

$$330°$$

Question

What is this angle in radians?

A. $\pi/6$
B. $11\pi/6$
C. $7\pi/6$
D. $5\pi/6$

Answer and reasoning

$330 \cdot \pi/180 = 330\pi/180 = 11\pi/6$, dividing by 30. 330° is 30° short of a full turn, and $11\pi/6$ is $\pi/6$ short of 2π. $\pi/6$ is 30°, the reference angle; $7\pi/6$ is 210°, the mirror image across the y-axis; and $5\pi/6$ is 150°, pointing the opposite way, half a turn from 330°. ***11π/6*.**

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Example 5: Two Thirds of Pi

$$2\pi/3$$

Question

What is this angle in degrees?

A. 60°
B. 300°
C. 120°
D. 270°

Answer and reasoning

$2\pi/3 \cdot 180/\pi = 360/3 = 120°$. Two thirds of a half turn is 120°. 60° is the reference angle, the distance to the x-axis; 300° is $360° - 60°$, the reference angle placed in Quadrant IV, which points the opposite way from 120°, half a turn away; and 270° flips the fraction to $3/2 \cdot 180$. **120°.**

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Example 6: Seven Fourths of Pi

$$7\pi/4$$

Question

What is this angle in degrees?

A. 315°
B. 45°
C. 225°
D. 135°

Answer and reasoning

$7\pi/4 \cdot 180/\pi = 7 \cdot 45 = 315°$. Each $\pi/4$ is 45°, and there are 7 of them. 45° is the reference angle; 225° is the mirror image across the y-axis, in Quadrant III; and 135° is the mirror image across the origin, in Quadrant II, half a turn away. **315°.**

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Example 7: A Third of Pi

$$\pi/3$$

Question

What is this angle in degrees?

A. 120°
B. 60°
C. about 1.05°
D. 300°

Answer and reasoning

$\pi/3 \cdot 180/\pi = 60°$, a third of a half turn. 120° takes π as 360°, a full turn instead of a half; about 1.05° writes the number $\pi/3 \approx 1.05$ with a degree sign, but that is the angle in radians, not degrees; and 300° is $360° - 60°$, which is coterminal with $-60°$, not with $60°$. **60°.**

Example 8: Seven Sixths of Pi

$$7\pi/6$$

Question

What is this angle in degrees?

A. 30°
B. 150°
C. 210°
D. 330°

Answer and reasoning

$7\pi/6 \cdot 180/\pi = 7 \cdot 30 = 210°$. Each $\pi/6$ is 30°. 30° is the reference angle; 150° is the mirror image across the x-axis, in Quadrant II; and 330° is $360° - 30°$, in Quadrant IV. **210°.**

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Example 9: Read the Angle

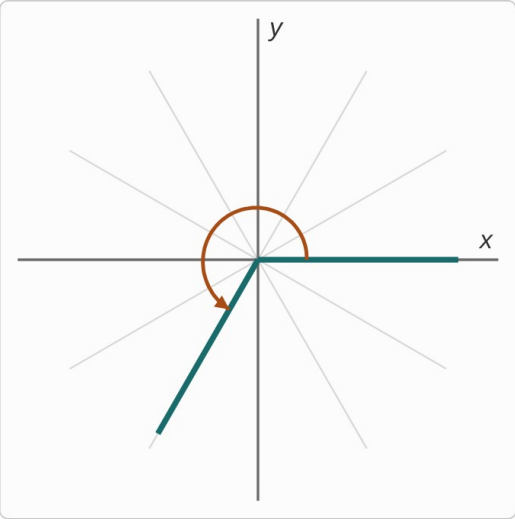


Figure: An angle in standard position, with faint guide lines every 30°. The arc turns counterclockwise from the positive x-axis, past the positive y-axis and the negative x-axis, and stops on the second guide line after the negative x-axis, below and to the left of the origin.

Question

The guide lines are 30° apart. What is the angle, in radians?

- **A.** $2\pi/3$
- **B.** $5\pi/3$
- **C.** $4\pi/3$
- **D.** $7\pi/6$

Answer and reasoning

The arc passes the negative x-axis (180°) and goes 2 guide lines farther, $180°+60°=240°$. In radians, $240\cdot\pi/180=4\pi/3$. $2\pi/3$ is 120°, the mirror image across the x-axis, above it instead of below; $5\pi/3$ is the mirror image across the y-axis; and $7\pi/6$ is 210°, one guide line short. ***$4\pi/3$*.**

Example 10: Which Quadrant

$$5\pi/4$$

Question

In which quadrant does the terminal side of this angle lie?

A. Quadrant I
B. Quadrant II
C. Quadrant III
D. Quadrant IV

Answer and reasoning

Compare with the landmarks: $\pi=4\pi/4$ and $3\pi/2=6\pi/4$. So $5\pi/4$ is between a half turn and three quarter turns, in Quadrant III, 45° past the negative x-axis. Quadrant I would need less than $\pi/2$; Quadrant II, between $\pi/2$ and π; and Quadrant IV, between $3\pi/2$ and 2π. **Quadrant III.**

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Example 11: On an Axis

$$\pi$$

Question

CheckIt writes this angle as π. Where is its terminal side in standard position?

A. Along the positive y-axis
B. Along the negative x-axis
C. Along the negative y-axis
D. Inside the second quadrant

Answer and reasoning

π radians is a half turn, 180°. Counterclockwise from the positive x-axis, a half turn points left, along the negative x-axis. An angle on an axis is in no quadrant. The positive y-axis is a quarter turn, $\pi/2$; the negative y-axis is three quarter turns, $3\pi/2$; and the second quadrant is strictly between $\pi/2$ and π. **Along the negative x-axis.**

Example 12: Describe the Turn

$$5\pi/3$$

Question

Which describes how to draw this angle in standard position?

A. Turn clockwise 300° from the positive x-axis
B. Turn counterclockwise 150° from the positive x-axis
C. Turn counterclockwise 300° from the positive x-axis
D. Turn clockwise 150° from the positive y-axis

Answer and reasoning

$5\pi/3 \cdot 180/\pi = 300°$. A positive angle in standard position starts on the positive x-axis and turns counterclockwise, so turn 300° counterclockwise, ending in Quadrant IV. Turning clockwise 300° would be the angle $-5\pi/3$; 150° takes π as 90°, $5/3 \cdot 90$; and clockwise 150° from the positive y-axis makes three slips: the direction, the size, and starting from the y-axis instead of the positive x-axis. **Turn counterclockwise 300° from the positive x-axis.**

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Example 13: Slip 1

$$240° \cdot 180/\pi = 43200/\pi$$

Question

A student converted 240° to radians. What is the mistake, if any?

A. There is no mistake.
B. Use π/180: 240π/180 = 4π/3.
C. Divide by 360 instead, which gives 240/360 = 2/3.
D. Use π = 3.14 first: 240 × 3.14 = 753.6.

Answer and reasoning

To go to radians, the degrees must cancel, so multiply by $\pi/180$: $240\pi/180 = 4\pi/3$. The student's $43200/\pi \approx 13{,}751$ radians would be over 2,000 full turns. Dividing by 360 gives $2/3$, the fraction of a full turn, not the angle in radians; and 240×3.14 substitutes 3.14 for π and never divides by 180. ***4π/3*.**

Example 14: Slip 2

$$5\pi/6 \cdot 180/\pi = 900/6 = 150°$$

Question

A student converted $5\pi/6$ to degrees. What is the mistake, if any?

A. There is no mistake.
B. Multiply by $\pi/180$ instead, to get $5\pi^2/1080$.
C. Degrees need π too: the answer is $150\pi°$.
D. $5\pi/6$ is in Quadrant III, so it is 210°.

Answer and reasoning

Multiplying by $180/\pi$ cancels the π: $(5 \cdot 180)/6 = 150°$. The student is right. Multiplying by $\pi/180$ goes the wrong way and leaves π^2; the π cancels here, so this answer in degrees has none; and $5\pi/6$ is less than π, so it is in Quadrant II, not III. **There is no mistake.**

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Example 15: Slip 3

$$315° \cdot \pi/360 = 7\pi/8$$

Question

A student converted 315° to radians. What is the mistake, if any?

A. There is no mistake.
B. The π should not be there: the answer is 7/8.
C. Use 180, not 360: 315π/180 = 7π/4.
D. Flip the fraction: the answer is 8π/7.

Answer and reasoning

π radians is 180°, not 360°, so the factor is $\pi/180$: $315\pi/180 = 7\pi/4$. A check: 315° is almost a full turn, and $7\pi/4$ is almost 2π, but $7\pi/8$ is less than a half turn. Radians for these angles keep the π; and the student's fraction is not upside down, it uses the wrong half-turn value. ***7π/4*.**

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Example 16: Slip 4

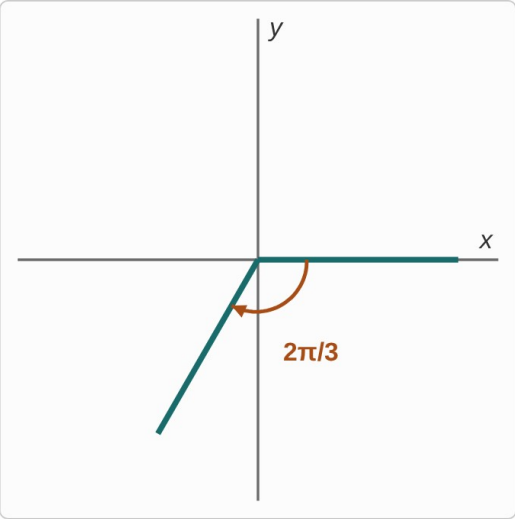


Figure: A student's drawing of an angle labeled 2π/3. The arc starts on the positive x-axis and turns clockwise, downward, ending with the terminal side below and to the left of the origin.

$$2\pi/3$$

Question

A student drew this angle in standard position. What is the mistake, if any?

A. There is no mistake.
B. The angle should start from the positive y-axis instead.
C. 2π/3 is more than half a turn, so it ends below the x-axis.
D. It should turn counterclockwise, into Quadrant II.

Answer and reasoning

$2\pi/3 = 120°$, a positive angle, so it turns counterclockwise from the positive x-axis and ends in Quadrant II, above the axis. The student turned 120° clockwise, which is the angle $-2\pi/3$. Standard position starts on the positive x-axis; and $2\pi/3$ is less than π, so it is less than half a turn. **Turn counterclockwise, into Quadrant II.**

Example 17: Slip 5

$$3\pi/4=(3(3.14))/4\approx 2.36°$$

Question

A student converted 3π/4 to degrees. What is the mistake, if any?

A. There is no mistake.
B. That is radians; times 180/π, it is 135°.
C. Use π = 3 instead, so the answer is 2.25°.
D. Multiply by π/180, so the answer is about 0.04°.

Answer and reasoning

About 2.36 is $3\pi/4$ as a number of radians, not degrees. To get degrees, multiply by $180/\pi$: $3\pi/4 \cdot 180/\pi = 3 \cdot 45 = 135°$. A different value for π still leaves the answer in radians; and multiplying by $\pi/180$ goes the wrong way. **135°.**

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